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Original datasets, R code for analysis and Rmarkdown code for creation of the manuscript are available at Zenodo [DOI WILL BE INSERTED HERE].

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Delayed seed germination to cope with environmental stress and disturbance in the European flowering plants

Running title: Delayed seed germination, stress and disturbance

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Summary

- Seeds of many plant species are programmed not to germinate under various environmental scenarios. Thus, delayed seed germination is widespread, despite the higher fitness expected for earlier germinating seeds. We explore delayed germination in the European flowering plants by testing the hypothesis that it is a mechanism to cope with stress and disturbance during regeneration.
- We retrieved 17,597 records representing 1,357 species from SeedArc, a global seed germination database. We classified species into four stress groups (low, drought, cold, wetlands) and two disturbance groups (low, high) using their ecological preferences for temperature, moisture and disturbance. We tested the likelihood of delayed germination as a function of stress-disturbance using phylogenetically informed meta-analysis.
- Delayed germination is more likely in species adapted to cold, and less likely in species adapted to drought and disturbance. Species adapted to low stress and low disturbance still present a degree of delayed germination. Different stress-disturbance groups respond differently to germination cues.
- Delayed germination is widespread in temperate flowering plants and works as a mechanism to cope with cold, disturbance and competition. The regeneration niche of the angiosperms shows a fundamental divide between those adapted to cold and those adapted to drought.

Keywords

Plant reproduction, plant regeneration, regeneration niche, seed dormancy, seed germination, delayed development, cold stress, drought stress

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Introduction

Developmental delay is a condition in which a developmental process is arrested even though the environment is favorable for its completion (Tuljapurkar, 1990; Tuljapurkar & Wiener, 2000). Developmental delay often results in slower growth rates and a delay in reaching the reproductive stage. But early development and rapid reproduction should be, in theory, associated with high individual fitness and thus favored by fecundity selection (Donohue *et al.*, 2010; Scott & Otto, 2014). Even so, developmental delay is common in many species across the tree of life (Tuljapurkar, 1990; Tuljapurkar & Wiener, 2000). A widespread example of developmental delay in plants is delayed seed germination, i.e. the situation in which a seed may not germinate even if exposed to moisture and temperature conditions favorable for post-germination survival and development (Scott & Otto, 2014). Delayed seed germination can result from two factors: inherent seed dormancy and environmentally imposed delayed seed germination (Nikolaeva *et al.*, 1985; Finch-Savage & Leubner-Metzger, 2006; Baskin & Baskin, 2014; Lamont & Pausas, 2023). Inherent seed dormancy is a state in which physicochemical properties of the seed prevent progress towards germination even if external conditions are otherwise suitable (Finch-Savage & Leubner-Metzger, 2006; Baskin & Baskin, 2014). Environmentally imposed delayed seed germination (referred to as 'imposed dormancy' in Lamont & Pausas (2023)) is a consequence of the external environmental cues not being suitable for triggering germination; these cues must be replaced by suitable ones for germination to start. Such cues include signals like specific levels and combinations of moisture, temperature, light and chemicals (Bewley *et al.*, 2013). These two kinds of factors (inherent and imposed) have been the subject of a long terminological debate in plant ecology and physiology (Vleeshouwers *et al.*, 1995; Lamont & Pausas, 2023) but, regardless of semantic issues, both of them provide mechanisms for the ecophysiological function of delayed seed germination. It is important to note that, in many plant species, the set of conditions allowing germination (i.e. the seed germination niche) is narrower than

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the set of conditions in which the species can grow and persist. In other words, the seeds of many species are programmed *not* to germinate under many environmental scenarios, meaning that delayed seed germination is widespread in the world's flora (Baskin & Baskin, 2014). This is counterintuitive given an expectation of increased fitness for seeds that are less dormant and germinate faster, resulting in more offspring and larger seedlings within each reproductive cycle (Pincheira-Donoso & Hunt, 2017).

Delayed seed germination has been understood as a regulator of regeneration phenology, integrating environmental cues from long-term seasonal variation (e.g. physiological dormancy release by overwintering) with short-term and microhabitat cues (e.g. light changes in the soil depth profile) (Finch-Savage & Leubner-Metzger, 2006). This regulation would ensure that seeds germinate in the right place at the right time, maximizing the probability of subsequent seedling survival and growth (Donohue *et al.*, 2010). In the plant regeneration process, delayed seed germination can only be adaptive in the presence of temporally heterogeneous regeneration environments with variable odds of seedling survival and growth (Pausas *et al.*, 2022). If the temporal occurrence of favorable regeneration environments is unpredictable (e.g. erratic rain events in a desert), the *'bet-hedging hypothesis'* proposes that delayed seed germination within a seed population can act to distribute germination events across different times, increasing the chance that at least part of the seeds produced by a given plant will germinate during a favorable timeframe conducive to life cycle completion (Venable & Brown, 1988; Gremer & Venable, 2014). Alternatively, if the temporal occurrence of favorable regeneration environments occurs regularly (e.g. snowmelt in an alpine snowbed), the *'best-bet hypothesis'* proposes that seeds can have specific germination requirements that track predictable environmental changes and therefore concentrate their germination in the most favorable regeneration window. To our knowledge, the term *'best-bet hypothesis'* was recently coined by Pausas *et al.* (2022), but its underlying logic has a long tradition in seed ecology, for example,

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in the interpretation of germination strategies in winter vs. summer annuals in response to predictable intra-annual seasonality (Baskin & Baskin, 1976). In winter annuals, individuals shed their mature seeds in spring, but those seeds delay their germination until autumn, so seedlings avoid predictable drought during summer (Baskin & Baskin, 1976). In this case, delayed seed germination is achieved by a requirement for summer dry-hot conditions to overcome physiological seed dormancy (Baskin & Baskin, 1976), plus an environmental delay caused by inadequate levels of temperature and moisture in the soil (Jiménez-Alfaro *et al.*, 2018), and these traits are understood to be essential for the winter annual life cycle in temperate seasonal environments (Footitt *et al.*, 2014).

To date, the macroecological patterns of delayed seed germination have been studied using macroclimate as a surrogate for regeneration environment favorability (Rubio de Casas *et al.*, 2017; Carta *et al.*, 2022a; Zhang *et al.*, 2022; Rosbakh *et al.*, 2023). These efforts have served to test and validate many aspects of the best-bet hypothesis as formulated by biogeographical seed ecology (Baskin & Baskin, 2014). However, as pointed out recently (Pausas *et al.*, 2022; Lamont & Pausas, 2023), a needed next step is to go beyond macroclimate and focus on the various aspects of the regeneration environment that determine its suitability and its relation to delayed seed germination, e.g. habitat features (Cruz-Tejada *et al.*, 2024). Since the major drivers of plant life histories are linked with stress and disturbance (Grime, 1979), we may expect that these are relevant factors to explain the favorability of regeneration environments. Grime (1979) defined stress as an external constraint to the rate of dry matter production in all or part of the vegetation. In the regeneration process, environmental stress has been described as the existence of a non-suitable season for germination and seedling establishment, resulting in suboptimal seedling growth or even mortality, and has usually been linked to macroclimatic patterns of frost and drought (Jurado & Flores, 2005). Plants adapted to frost- or drought-limited ecosystems should have germination traits that ensure that seed germination is delayed during the cold or dry season, respectively (Jurado & Flores, 2005; Baskin

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135 & Baskin, 2014). Examples for temperate regions include alpine plants with seeds
136 whose physiological dormancy prevents germination during autumn and winter
137 (Fernández-Pascual *et al.*, 2021; Rosbakh *et al.*, 2023); and Mediterranean species
138 with cold-cued germination outside the summer dry season (Doussi & Thanos, 2002;
139 Carta *et al.*, 2022a; Cruz-Tejada *et al.*, 2024). Independently from stress,
140 disturbance has been defined as a measurable loss of biomass in a biological
141 community (Grime, 1979; White & Jentsch, 2001). The relationship of disturbance
142 with delayed seed germination has been particularly studied in fire-prone
143 ecosystems (Baker *et al.*, 2005; Dairel & Fidelis, 2020). In these systems, fire can
144 act both as a cue for dormancy release (smoke, heat) and as a creator for favorable
145 regeneration conditions (low competition, high resource availability, low predation,
146 low pathogen load) (Pausas *et al.*, 2022). Furthermore, ecological theory
147 (MacArthur, 1962; MacArthur & Wilson, 1967) also predicts that frequent disturbance
148 should select short-lived annual plants producing many small seeds with a high
149 capacity for dispersal in space (Venable & Brown, 1988), for example, ruderal annual
150 plants adapted to disturbed anthropogenic habitats. Given the trade-off between
151 seed dispersal in space and seed dispersal in time via delayed seed germination
152 (Chen *et al.*, 2020), we would expect highly disturbed environments to be associated
153 with low inherent germination delay (i.e. low seed dormancy), as plants would
154 escape from high disturbance by producing many low-cost and highly dispersible
155 seeds with high germinability, to maximize the chance of germinating in a place with
156 a favorable situation along the disturbance cycle (but this assumes that a mosaic of
157 disturbance-cycle stages occur simultaneously in the landscape).

158 Here, we explore the patterns of delayed seed germination in response to stress and
159 disturbance, the underlying drivers of regeneration environment suitability. We
160 perform a meta-analysis of primary germination records using a large dataset
161 representing > 17,000 records of > 1,300 species from 97 families of the European
162 flowering plants. Rather than measuring delayed seed germination as the
163 presence/absence of seed dormancy (Rubio de Casas *et al.*, 2017; Zhang *et al.*,

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164 2022; Rosbakh *et al.*, 2023), our meta-analysis considers the multi-faceted
165 responses of seed germination to various cues mapping onto inherent (scarification,
166 cold stratification) and externally imposed (temperature regimes and light)
167 germination delays (Carta *et al.*, 2022a). Based on the hypothesis that delayed seed
168 germination is a mechanism to cope with stress and disturbance during the
169 regeneration phase, we test the following predictions: (1) delayed seed germination
170 is more likely in species adapted to high stress and/or low disturbance; (2) specific
171 germination cues driving delayed seed germination vary depending on species
172 preferences for stress and disturbance following the expectations developed in
173 **Table 1**.

174 *Table 1: Environmental cues driving delayed seed germination in the European*
175 *flowering plants. The table lists the cues included in our meta-analysis. For each*
176 *cue, the table provides a description of the laboratory experimental treatment used*
177 *as a proxy of the environmental cue, the theorized ecological function of the cue in*
178 *regulating germination phenology, and our prediction in regard to the relationship*
179 *between the cue and species preferences for stress and disturbance.*

GERMINATION CUE	EXPERIMENTAL TREATMENT	ECOLOGICAL FUNCTION	PREDICTION
Scarification	Abrasion of the seed coat was applied to allow water imbibition; binary (yes/no)	Delay germination until seed coat is abraded by soil movement, fire, freeze-thaw, etc.	Stronger signal for high-disturbance species (e.g. germinate after fire)
Cold stratification	Seeds underwent a period (> 4 weeks) of cold-wet stratification (< 6ºC); binary (yes/no)	Delay germination until a cold season is passed	Stronger signal for cold-stress species (e.g. avoid autumn-winter germination)
Average temperature	Weighted average of the diurnal temperature cycle, weighted by the duration of each phase; in ºC	Delay germination until the proper time of the annual temperature cycle	Stronger signal for cold-stress/low-disturbance species (e.g. summer germination in

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GERMINATION CUE	EXPERIMENTAL TREATMENT	ECOLOGICAL FUNCTION	PREDICTION
			alpine meadows, mires)
Alternating temperature	Different temperatures (>5ºC) were applied during the day and night, in diurnal cycles; binary (yes/no)	Delay germination unless close to soil surface, under open vegetation and/or a low water table	Stronger signal for low-stress species (e.g. germinate in vegetation gaps)
Light	Seeds were exposed to light during some part of the diurnal cycle; binary (yes/no)	Delay germination unless close to soil surface and/or under open vegetation	Stronger signal for low-stress species (e.g. germinate in vegetation gaps)

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Materials and Methods

Seed germination dataset

We prepared a curated pan-European seed germination dataset by retrieving records from *SeedArc*, the global archive of primary seed germination data (Fernández-Pascual *et al.*, 2023). We define a record as a germination proportion of a given seed lot of a species, recorded in response to a given set of germination cues recreated in laboratory experimental conditions. To ensure data quality, we only retrieved records belonging to:

1. Angiosperms, to avoid the effect of gymnosperm branch lengths on subsequent phylogenetic analysis.
2. Species present in the megatree of the seed plants by Smith & Brown (2018), a tree that would be used in subsequent analysis.
3. Experiments conducted with seeds collected from wild populations in Europe and neighboring regions, between 30º W and 70º E and north of 30º N.

4. Laboratory experiments conducted with either agar or filter paper as substrate.
5. Experiments not using specialized and uncommon treatments (e.g. sterilization, nitrates, plant hormones, UV light).
6. Experiments not using complex dormancy breaking cycles with warm stratification or several steps (e.g. move-along experiments), which would be incomparable with simpler and more common experiments.
7. Experiments conducted with at least 5 but no more than 1,000 seeds per experimental replicate.
8. Species with data available for the ecological indicator values (see below).

Furthermore, to merge the heterogeneous original records, we simplified the germination cues by merging the originally recorded treatment levels into simpler treatment levels (described in **Table 1**) that are routinely recorded and comparable across germination tests. These cues are proxies of underlying quantitative variables that drive the physiological responses of seeds, e.g. the heat thresholds for scarification, the length and temperature of cold stratification, the cardinal germination temperatures, the amplitude of the diurnal thermal oscillations, the length of the photoperiod, etc. Further, to merge and homogenize this dataset with the stress-disturbance dataset (see below), we standardized all species names using the World Checklist of Vascular Plants V.10 (Govaerts *et al.*, 2021).

Stress and disturbance groups

We classified species in the germination dataset based on their preferences for stress and disturbance. To create this classification, we used the species' ecological indicator values (EIVs) for temperature, moisture, disturbance frequency and disturbance severity (Dengler *et al.*, 2023; Midolo *et al.*, 2023; Tichý *et al.*, 2023). Although these indicators were developed for adult plants, the ecological preferences of adult plants should be reliable proxies of the environmental conditions

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221 in which their seeds germinate. EIVs are a numerical ordering of species according
222 to their optima along environmental gradients. For temperature and moisture, we
223 compared the EIVs provided by Dengler *et al.* (2023) and Tichý *et al.* (2023) and
224 verified that (1) their values were comparable and (2) the selection of one or the
225 other did not affect subsequent analysis. Thus, we decided to use the values by
226 Dengler *et al.* (2023) because they covered more of the species in the germination
227 dataset. For the disturbance EIVs, we used the values of disturbance frequency
228 ($\log_{10}$ of the inverse of the estimated mean interval between two consecutive
229 disturbance events, in centuries) and disturbance severity (a proportional value, 0/1,
230 of biomass loss in the community during a disturbance event) provided by Midolo *et*
231 *al.* (2023). In the case of the disturbance indicators, we decided to use the version
232 of the EIVs calculated for the herb layer, since it is in the herb layer where
233 regeneration by seed takes place.

234 To create the classification, we treated temperature and moisture separately, to
235 reflect the separate effects of cold and drought stress (**Fig. 1A**). As a first step, we
236 classified each species for each of the two indicators, temperature and moisture.
237 The EIVs of Dengler *et al.* (2023) are continuous and take values from 0 to 10.
238 Therefore, we classified species into 'low', 'mid' or 'high' temperature and moisture
239 categories if the value of the respective indicator was below 3.3, between 3.3 and
240 6.7, or above 6.7, respectively. If a species was in the 'low' temperature range, we
241 assigned it to the 'cold stress' group, assuming that its habitats are typically
242 characterized by an unfavorable cold season (e.g. with winter frost events) strong
243 enough to affect seedling regeneration. Correspondingly, if a species was in the 'low'
244 moisture range, we assigned it to the 'drought stress' group, assuming that its
245 habitats are typically affected by an unfavorable dry season (e.g. with summer
246 drought events) similarly strong enough to affect seedling regeneration. There were
247 23 species that had 'low' values for temperature and moisture; these were species
248 from southern European Mediterranean mountains, and we classified them into the
249 'cold stress' group assuming that cold temperatures would be the major unfavorable

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factor during their regeneration, in accordance with the view that the seed germination of Mediterranean mountain species is relatively similar to that of temperate alpine species (Giménez-Benavides *et al.*, 2005). Another 22 species had 'low' temperature and 'high' moisture indicators; these were species from Arctic-Alpine bogs and fens, and we also classified them into the 'cold stress' group. The species in the 'mid' range for temperature and moisture went into the 'low stress' group. We also classified as 'low stress' a small group of 27 species that were in the 'high' temperature range but had 'mid' moisture values, assuming that in the European context warm temperatures should not be unfavorable for regeneration if moisture is available. Finally, we created a 'wetland' group for the species with 'mid' temperature and 'high' moisture requirements; these are hydrophytes and helophytes adapted to life in water or flooded substrates. In these habitats, soil waterlogging is primarily dependent on edaphic factors, and hence, they do not fit into the general framework of seasonal unfavorable regeneration environments created by cold or drought stress. There were no species with 'high' values for both temperature and moisture.

The indicators of disturbance frequency and severity were strongly and positively correlated (Pearson's $r > 0.7$). This correlation suggested that the two disturbance indicators could be merged into one disturbance factor with two levels (i.e. low and high; **Fig. 1B**). Consequently, we classified a species as 'low disturbance' if its EIVs for disturbance frequency and severity were below the median value of the EIV (< 1.7 frequency and 0.4 severity), i.e. if the species avoided frequent or severe disturbances; and we classified a species as 'high disturbance' if it did not meet the previous condition.

Therefore, our final classification included four stress groups (low stress, drought stress, cold stress and wetlands) and two disturbance groups (low and high), resulting in 8 stress - disturbance groups (**Fig. 1C**). It must be noted that our high stress-high disturbance groups do not correspond to Grime's 'missing' fourth corner (Herben *et al.*, 2018), as our stress groups reflect the favorability of the regeneration

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environment, rather than productivity. A Principal Component Analysis (PCA) of the indicator values (**Fig. 1D**) supported that our eight groups describe the ecological variability in the studied flora. One axis of variation separated species adapted to low disturbance from species adapted to high disturbance, with disturbance frequency and severity showing high covariation. A second axis, orthogonal to the disturbance axis, ordered species according to their stress group: the central space of this axis was occupied by low stress species, while the two extremes were occupied by species adapted to cold or drought stress. The wetland species were grouped close to the cold stress species.

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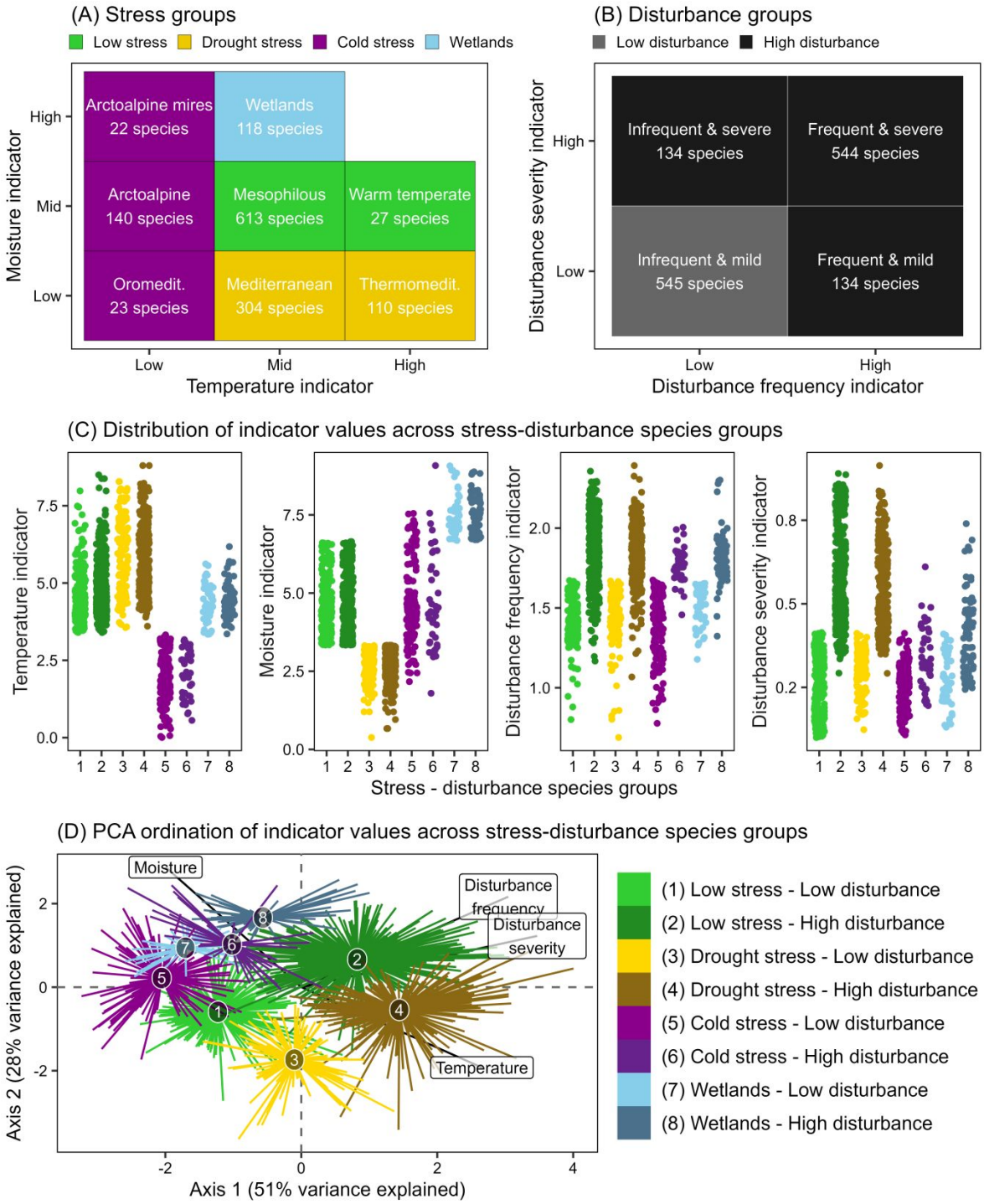

Figure 1: Stress-disturbance groups. (A) Combinations of temperature and moisture into four stress groups (low stress, drought stress, cold stress and wetlands). (B) Combinations of disturbance frequency and severity into two disturbance groups

(low and high). (C) Distribution of indicator values (temperature, moisture, disturbance frequency and disturbance severity) in each of the eight stress-disturbance groups. (D) Principal Component Analysis (PCA) biplot of the indicator values, with species grouped and colored according to their stress-disturbance group. The labeled arrows indicate the contribution of the indicator values to the PCA axes.

Statistical analysis

We tested the variation in delayed seed germination among stress-disturbance groups by performing a meta-analysis of the germination dataset using binomial generalized mixed models with Bayesian estimation (Markov Chain Monte Carlo generalized linear mixed models, MCMCglmms) as implemented in the R package *MCMCglmm* (Hadfield, 2010). A detailed description of the MCMCglmm methodology in the context of seed germination meta-analysis is provided by Carta *et al.* (2022a). In this analysis, final germination proportion, i.e. the number of seeds germinated out of the number of viable seeds in the experiment, represents the model response variable. Since in our study the germination test conditions (i.e. constant moisture supply and 95% of the tests in the temperature range 7-26 ºC) were generally adequate for seedling survival and growth, we assumed that any significant deviation from 100% final germination represented a signal of delayed seed germination.

To test our first prediction that delayed seed germination should be more likely in species adapted to high stress and low disturbance, we fitted a model across the whole dataset, with two predictors: stress (four levels; low, drought stress, cold stress, wetlands) and disturbance (two levels; low and high) (see model syntax in **Appendix S1**). The low stress - low disturbance group was the reference level to which all remaining groups were compared. We calculated models with and without a stress x disturbance interaction. As the interaction was not significant, in the following, presentation and interpretation of results are based on the model without interaction.

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To test our second prediction that the specific germination cues driving delayed seed germination should vary depending on the species preferences for stress and disturbance, we fitted separate models for subsets of the dataset representing each of the eight stress-disturbance groups, with germination cues (scarification, cold stratification, average temperature, alternating temperature and light) as predictors (see model syntax in **Appendix S1**).

Predictor variables were centered and scaled so their contribution to the effect sizes could be compared. We used weakly informative priors, with parameter-expanded priors for the random effects (Hadfield, 2010). In all models, random effects included (1) the source of the germination data (to account for between-study variation related to the research group or publication from which the data was extracted), (2) seed lot ID (to account for between-study variation related to differences in experimental settings, seed lot quality and storage conditions, etc.), (3) species identity (to account for between-study variation related to differences between study species' traits) and (4) a reconstructed phylogenetic tree for the study species (to account for shared phylogeny). To create the phylogeny we used the R package *U.PhyloMaker* (Jin & Qian, 2023) which contains an updated mega-tree of the seed plants based on Smith & Brown (2018). In addition, to take into account the potential effect of seed mass on germination responses and species ecologies, we repeated the models with seed mass as a covariable, using the seed mass dataset curated by Carta *et el.* (2022b). The values of seed mass did not differ between stress-disturbance groups, and the inclusion of seed mass did not affect model outputs for the study variables. Therefore, we kept seed mass out of the analyses for simplicity.

Each model was run for 500,000 MCMC steps, with an initial burn-in phase of 50,000 and a thinning interval of 50 (De Villemereuil & Nakagawa, 2014), resulting, on average, in 9,000 posterior distributions. From the resulting posterior distributions, we calculated mean parameter estimates and 95% highest posterior density and credible intervals (CI). We interpreted the significance of model parameters by examining CIs, considering parameters with CIs overlapping with zero as non-

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350 significant. We performed all data analyses with R version 4.3.1 (R Core Team,
351 2023), using the R package 'tidyverse' (Wickham *et al.*, 2019) for data processing
352 and visualization. The original datasets, as well as R code for analysis and creation
353 of the manuscript, can be accessed at Zenodo (see Data Availability Statement).

354 **Results**

355 *Representativeness of the germination dataset*

356 The final germination dataset contained 17,597 experimental records obtained from
357 187 sources (i.e. research groups providing their primary germination data or
358 published references from which data had been extracted). Experiments had been
359 conducted with a total of 795,345 seeds from 5,009 seed collections originating from
360 Europe and neighboring regions (**Appendix S2**). Seeds had been collected across
361 the study area, with 41 countries represented in the dataset. The dataset included
362 1,357 species from 97 angiosperm families and 32 orders, representing the major
363 clades of the angiosperms (**Appendix S2**). The dataset represented all the major
364 European habitat types, considering both the numbers of species (**Appendix S2**)
365 and the number of records (**Appendix S2**). Most species and records came from
366 zonal and non-anthropogenic habitat types (i.e. forests, shrublands and grasslands;
367 **Appendix S2**). Experiments had used scarified seeds in 3,393 records (19%) and
368 cold-stratified seeds in 3,019 records (17%). The average temperatures of the
369 experiments ranged from 0 to 40 ºC, with an average of 18 ºC, and 95% of the tests
370 in the temperature range 7-26 ºC. Alternating temperatures had been used in 7,377
371 records (42%) and dark conditions in 1,490 records (9%).

372 *Testing prediction 1*

373 Regarding our prediction that delayed seed germination should be more likely in
374 species adapted to high stress and low disturbance, we found that, independent of
375 experimental conditions (i.e. across germination cues), species adapted to drought
376 stress had significantly higher final germination than low-stress species, while

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species adapted to cold stress had significantly lower germination (**Fig. 2**, **Appendix S1**). In ecological terms, these findings imply that species of water-limited habitats tend to germinate readily across several experimental treatments and thus do not show delayed seed germination, while species from temperature-limited ecosystems are more likely to show delayed seed germination. Wetland species behave similarly to cold stress species, but their credible interval slightly overlaps with zero (our criterion for no statistical significance), possibly because this group had fewer records, resulting in lower statistical power. Finally, high-disturbance species had significantly higher overall germination, i.e. species that typically occur in high-disturbance habitats tend to germinate more readily across several experimental treatments, and thus are less likely to display delayed seed germination.

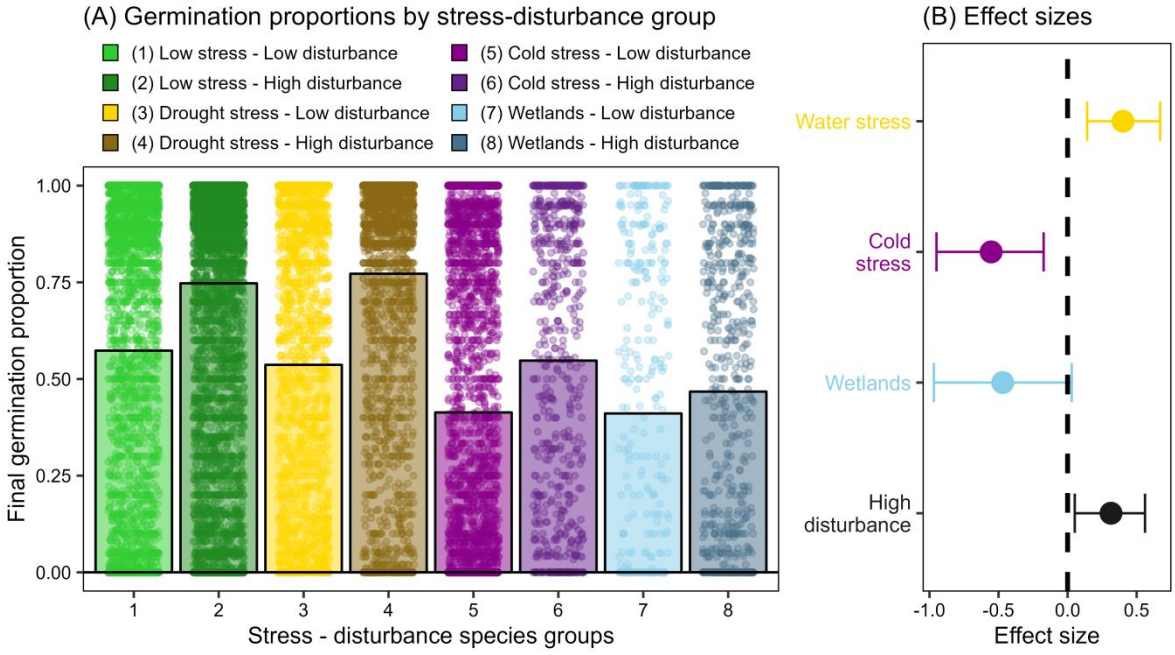

Figure 2: Seed germination across stress-disturbance groups. (A) Final germination proportions in each of the eight stress-disturbance groups, across all germination records. The dots indicate the final germination proportion in each of the original germination records, and the bars indicate the average proportion per group. (B) Effect sizes of the MCMCglmm testing the effect of stress and disturbance on final germination proportion. The reference level is the low stress - low disturbance group.

Dots indicate the posterior mean of the effect size for each stress group and the high disturbance group, and whiskers the 95% credible interval of the effect size. The line of zero-effect is shown: when a credible interval overlaps with the zero-effect line, the effect can be regarded as non-significant. Non-overlapping effects to the left of the zero-effect line indicate that the group has lower seed germination than the reference group, while non-overlapping effects to the left of the zero-effect line indicate that the group has higher seed germination than the reference group.

Testing prediction 2

Regarding our prediction that the specific germination cues driving delayed seed germination should vary depending on the species preferences for stress and disturbance (**Table 1**), we found that most stress-disturbance groups responded positively to most germination cues (**Fig. 3**, **Appendix S1**). That is, independently from stress-disturbance preferences, the European species in our dataset were more likely to germinate after scarification, after cold stratification, at higher and alternating temperatures, and with light. However, the effect sizes of each cue varied among stress-disturbance groups, as shown by non-overlapping credible intervals. Specifically, there were several exceptions to the general pattern:

1. No response to scarification was found in two groups: low stress - low disturbance and wetlands - low stress.

2. The positive response to cold stratification was stronger in the low disturbance than in the high disturbance groups, and it was negative in the drought stress - high disturbance group.

3. Average temperature had the strongest positive effect on the cold stress - low disturbance group, and was not significant in the two drought stress groups and in the wetlands - low disturbance group.

4. Alternating temperature only had a positive effect on the low stress and wetland groups, independently of disturbance level.

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5. Light only had a positive effect on the low stress groups and in the cold stress - low disturbance group.

In general, these results agreed with our predictions (**Table 1**) with some exceptions: (i) scarification had a strong effect in the groups of species adapted to high disturbance, but also in species adapted to low disturbance and high stress (drought and cold); (ii) cold stratification had a strong effect on species adapted to cold stress, but also on species from the low stress - low disturbance group; and (iii) alternating temperature and light had a strong effect on species adapted to low stress, but also on species adapted to cold stress or wetlands.

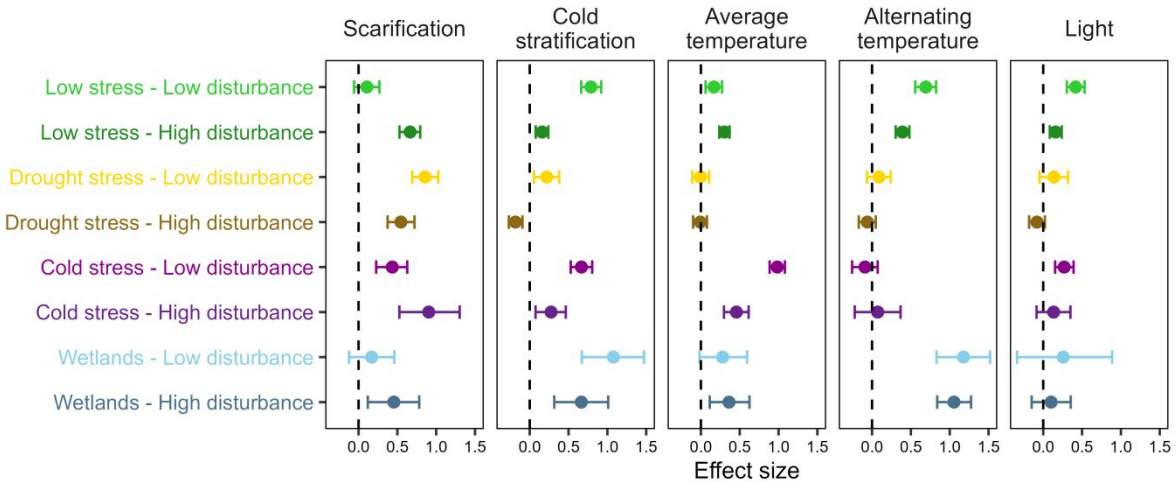

Figure 3: Germination cues driving delayed seed germination across stress-disturbance groups. Effect of germination cues (scarification, stratification, average temperature, alternating temperature and light) over the final germination proportions. Dots indicate the posterior mean of the effect size for each cue, and whiskers the 95% credible interval of the effect size. The line of zero-effect is shown: when a credible interval overlaps with the zero-effect line, the effect can be regarded as non-significant. Non-overlapping effects to the left of the zero-effect line indicate that the cue depresses seed germination, while non-overlapping effects to the left of the zero-effect line indicate that the cue promotes seed germination. In separate

panels, the figure shows the effect of the germination cues in each of the eight stress-disturbance models.

Random factors and phylogenetic effect

In the full model that included the entire germination dataset (i.e. the model testing the first prediction), the random factor with the strongest effect was phylogeny (mean = 3.56, CI = 2.36 - 4.82), followed by data source (mean = 2.48, CI = 1.61 - 3.45), seed collection (mean = 2.03, CI = 1.82 - 2.25) and species (mean = 1.28, CI = 0.98 - 1.62) (**Appendix S1**). In the models for the stress-disturbance groups (i.e. the models for the second prediction), the pattern was similar (**Appendix S1**).

Discussion

Our results indicate that delayed seed germination, by which a seed does not germinate even in a potentially favorable environment, is more likely in species adapted to low disturbance and cold stress, but less likely in species adapted to drought stress. In ecological terms, this suggests that delayed seed germination is less advantageous in species adapted to disturbed or water-limited ecosystems. These results agree partially with our first prediction (i.e. delayed seed germination should be more likely in species adapted to high stress and low disturbance), as we expected delayed seed germination to be more likely in the drought stress group, and we found some delayed germination in the low stress group. This suggests that in water-limited ecosystems, the delay in germination is externally regulated: in such systems, no rainfall means no soil moisture and no germination; therefore, there is no strong need to regulate germination delay by inherent seed properties. It also suggests that delayed seed germination can be a feature of competitive species adapted to mild and undisturbed environments.

Delayed seed germination is more likely in species adapted to cold stress, but not in species adapted to drought stress

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We show that delayed seed germination is a feature of cold adapted species, in agreement with the described germination strategies of alpine species, which delay germination until spring thanks to a combination of cold stratification requirements and warm-cued germination (Fernández-Pascual *et al.*, 2021). At the same time, our models indicate that delayed seed germination is less likely in species from water-limited ecosystems, e.g. Mediterranean zonal habitats. Although not initially contemplated by our predictions, this result aligns with the ecological features of dry ecosystems, where delayed seed germination can be achieved by external environmental factors (i.e. environmentally imposed delayed seed germination) (Lamont & Pausas, 2023). For example, regeneration phenology can be driven by the timing and intensity of rainfall events, which regulate soil water potential and create windows for seed germination (Carta *et al.*, 2013; Jiménez-Alfaro *et al.*, 2018). Since our dataset comprises laboratory experiments where water is provided to saturation, we cannot assess the effect of this key driver of germination in water-limited ecosystems. The scarcity of data on germination responses to water, compared to the amount of research dedicated to germination responses to temperature, suggests that a concerted effort is needed to explore seed regeneration in arid ecosystems. This becomes especially important considering that our results point to a fundamental divide in regeneration between angiosperms adapted to cold and those adapted to drought.

Delayed seed germination is less likely in species adapted to disturbance

Disturbance must be interpreted from the seed's point of view. Being relatively small and well defended, seeds are not necessarily affected by disturbance in the same way as adult plants are, i.e. by losing their aerial biomass or dying. For seeds, a disturbance represents an opportunity for seedling establishment, for example by reducing competition from the established vegetation. In this sense, the importance of disturbance as a gap provider depends on how well the frequency and severity of disturbance align with the life spans of the seed and its potential competitors (Keeley *et al.*, 2011; Pausas & Keeley, 2014). A systematic computer simulation of life history

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strategies using temperate grasslands as a model (Schippers *et al.*, 2001) concluded that delayed germination is superior to dispersal in space as a method for coping with disturbance. Our results contradict such conclusion. Our evaluation, based on a broader spectrum of disturbance regimes considering a sample of the European flora from different habitats and life forms, concludes that high disturbance leads to lower delayed seed germination, i.e. to lower dormancy and higher germinability. When disturbances are sufficiently frequent or severe to provide frequent or large regeneration gaps, the production of highly germinable seeds is favored, possibly as a way to occupy the more readily available regeneration gaps. It is to be remarked that, in terms of delayed seed germination, disturbance species behave similarly to drought stress species. Given that many of the disturbance-adapted synanthropic plants of Europe (e.g. ruderal plants, segetal plants, and species of trampled soils) have a Mediterranean and thermophilous origin (Zohary, 1950; Preston *et al.*, 2004), it can be hypothesized that disturbance-adapted germination strategies are a pre-adaptation to regeneration in water-limited ecosystems. Furthermore, disturbed bare soils dry out faster, suggesting an additional link between the functional responses to drought and disturbance.

Some degree of delayed seed germination is found in species adapted to low stress and disturbance

Our results indicate that delayed germination is widespread in the European flora, as we found a degree of delayed germination even in the low stress - low disturbance group (i.e. species which are infrequent in habitats characterized by cold stress, drought stress or high disturbance). Species in this low stress - low disturbance group are adapted to undisturbed and climatically mild ecosystems (e.g. broadleaved deciduous forests), where competition is high, and factors such as light may become the major drivers of plant strategies. It would seem that these competitive conditions also favor delayed seed germination. When conditions are stable and competition is high, the probability of a seed finding an available spatial gap is relatively low, as they are virtually non-existent (every empty gap/niche would

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be quickly preempted by the extant vegetation). Therefore, the chances for seed regeneration are only high when a temporal gap becomes available, e.g. a forest clearing after a windthrow. Also, in forest understories, seedling mortality could also result from continuous low light levels driving a photosynthetic deficit which, although exerted by extant canopy trees could, at the scale of the seedling, be interpreted as a stress factor (Blandino *et al.*, 2022). In any case, we need to consider that our study only includes European angiosperms. The vegetation history of Europe (e.g. glaciation cycles, seasonal climate, long history of synanthropic flora and vegetation) has shaped the continental species pool that harbors a high proportion of species regularly experiencing stress and/or disturbance in one form or another. Therefore, to fully test delayed seed germination with relation to stress, we should include in our analysis floras adapted to truly stable and mild regeneration environments (e.g. tropical rainforests). Unfortunately, such floras are currently not adequately represented in seed germination data (Visscher *et al.*, 2022; Fernández-Pascual *et al.*, 2023).

Delayed seed germination is achieved through different combinations of environmental cues, depending on species preferences for stress and disturbance

Our models support the prediction that different levels of stress and disturbance favor different responses to germination cues as drivers of delayed seed germination and germination phenology, but with some deviations from our initial expectations (**Table 1**). A strong response to scarification (i.e. a signal to remove an impermeable seed coat and physical dormancy (Baskin & Baskin, 2014; Lamont & Pausas, 2023)) was found to be more likely in species adapted to high disturbance or drought/cold stress. This supports the role of physical dormancy as a way to detect and respond to disturbance events such as fire (Ooi *et al.*, 2014; Pausas & Lamont, 2022) but also to delay germination during unfavorable seasons such as dry summers (Van Assche *et al.*, 2003; Van Assche & Vandelook, 2010; Luna *et al.*, 2022). The response to cold stratification (i.e. a signal to alleviate physiological dormancy (Finch-Savage & Leubner-Metzger, 2006)) was strong in species adapted to low

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disturbance and low stress, as well as in species adapted to cold stress. A positive response to cold stratification in species growing in habitats with an unsuitable cold season for germination (i.e. risk-avoidance of autumn-winter germination) agrees with previous findings (Jurado & Flores, 2005; Rubio de Casas *et al.*, 2017; Rosbakh *et al.*, 2023). In addition, our results suggest that in environments without a pronounced suboptimal season for seedling establishment, physiological dormancy as a means to delay germination can still be advantageous, possibly enabling species to avoid competition until gaps in the vegetation have formed. For example, in highly competitive deciduous forest vegetation, delayed seed germination in spring would be advantageous, as this might be the only window-of-opportunity for establishment within an otherwise closed forest canopy (Blandino *et al.*, 2022). Average germination temperature (i.e. an environmental signal of the annual temperature cycle (Rubio de Casas *et al.*, 2017; Fernández-Pascual *et al.*, 2021; Carta *et al.*, 2022a)) had a positive effect on most groups (preferring relatively warm germination temperatures) with the exception of drought-stress species. Taken together with the effect of cold stratification, these results reflect the fact that most of the studied European species are spring germinators (Baskin & Baskin, 2014). The effect of temperature is stronger (i.e. the preferred germination temperature is even warmer) in species from cold stress-low disturbance habitats, which agrees with the warm-cued germination reported for habitats such as alpine grasslands (Fernández-Pascual *et al.*, 2021). The lack of an effect of average temperature in drought stress species also agrees with the Mediterranean seed germination syndrome, characterized by cold-cued germination (Carta *et al.*, 2022a; Cruz-Tejada *et al.*, 2024). Finally, alternating temperatures and light are usually interpreted as interchangeable signals of microhabitat factors such as burial depth, canopy closure or water table height (Thompson & Grime, 1983; Pons, 2000; Carta *et al.*, 2017). Our results support this idea, as the effect of these two cues was strong in species from the low stress-low disturbance group, i.e. species adapted to habitats without a marked unfavorable regeneration environment, where microhabitat detection can be most important, signaling favorable gaps such as a forest clearing after tree fall.

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584 *Conclusions and future perspectives*

585 In conclusion, this study shows that delayed seed germination is more likely in
586 species adapted to cold stress or low disturbance. In species adapted to low
587 disturbance, delayed seed germination seems to be the norm, even in stable
588 environments without a marked unfavorable season for seed regeneration, except if
589 the ecosystem is drought prone. This highlights the potential role of competition
590 avoidance in developing delayed seed germination strategies. It also suggests that
591 there might be habitat specific factors driving the degree of delayed seed
592 germination. In this work, we have furthered the traditional understanding that
593 delayed seed germination results from unfavorable regeneration environments, by
594 deconstructing the concept of regeneration unfavourability along axes of stress and
595 disturbance. Future studies will need to extend the analysis to other floras, improving
596 our knowledge of tropical and arid biomes, and producing curated data from
597 macroecological experiments that carefully control for seed and adult environments,
598 species life forms and functional traits, habitats (e.g. forests vs. shrublands
599 vs. grasslands vs. wetlands) and separate stress factors (e.g. temperature,
600 moisture, light, salinity). Such experiments will continue to shed light on delayed
601 seed germination, which appears to be a widespread functional feature of the seed
602 plants.

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608 **Competing interests**

609 None declared.

Author contributions

Eduardo Fernández-Pascual: Conceptualization; Methodology; Investigation; Data Curation; Formal Analysis; Visualization; Writing – Original Draft Preparation; Writing – Review & Editing. Angelino Carta: Conceptualization; Methodology; Investigation; Data Curation; Writing – Review & Editing. Sergey Rosbakh: Conceptualization; Investigation; Writing – Review & Editing. Borja Jiménez-Alfaro: Conceptualization; Writing – Review & Editing. Diana María Cruz-Tejada: Investigation; Writing – Review & Editing. Markus Wagner: Investigation; Writing – Review & Editing. Lorenzo Pinzani: Investigation; Writing – Review & Editing. Katalin Lukács: Investigation; Writing – Review & Editing. Shyam S. Phartyal: Investigation; Writing – Review & Editing.

Data availability

Original datasets, R code for analysis and Rmarkdown code for creation of the manuscript are available at Zenodo [DOI WILL BE INSERTED HERE].

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787 **Supporting Information**

788 **Appendix S1** Model syntax and detailed output summaries for the meta-analysis
789 models.

790 **Appendix S2** Geographical, phylogenetic and habitat representativeness of the
791 seed germination dataset.

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